

The Bochner-Weitzenböck Formula

$$\frac{1}{2}\Delta(|\nabla f|^2) = \langle \nabla f, \nabla(\Delta f) \rangle + \|\nabla^2(f)\|^2 + \text{Ric}(\nabla f, \nabla f)$$

The Carré du Champ Operator (Γ)

$$\Gamma(f, h) = \frac{1}{2}(L(fh) - fL(h) - hL(f))$$

$$\Gamma(f, f) = \frac{1}{2}(L(f^2) - 2fL(f))$$

$$\Gamma(f, f) = |\nabla f|^2$$

The Iterated Carré du Champ (Γ_2)

$$\Gamma_2(f, f) = \frac{1}{2} \Delta \Gamma(f, f) - \Gamma(f, \Delta f)$$

Bochner-Lichnerowicz-Weitzenböck Identity

$$\Gamma_2(f, f) = \|\nabla^2(f)\|^2 + \text{Ric}(\nabla f, \nabla f)$$

The Weitzenböck Identity for Vector Fields

$$\frac{1}{2}\Delta(|V|^2) = \langle V, \Delta_{\text{Hodge}} V \rangle - |\nabla V|^2 - \text{Ric}(V, V)$$

$$\Gamma(V, V) = |\nabla V|^2$$

Vector Carré du Champ as Frobenius Norm

$$\Gamma(V, V) = \sum_{a=1}^D \Gamma(V_a, V_a)$$

$$\Gamma(V, V) = \sum_{a=1}^D |\nabla V_a|^2$$

$$|\nabla V|_{\text{Frobenius}}^2 = \sum_{a=1}^D |\nabla V_a|^2$$

$$\Gamma(V, V) = |\nabla V|^2$$

Markov Triples and Discrete Diffusion

$$(E, \mu, \Gamma)$$

$$L = P - I$$

$$\Gamma(f, f)(x_i) = \frac{1}{2} \sum_j P_{ij} (f(x_j) - f(x_i))^2$$

Calculating P (Coifman-Lafon Normalisation)

$$W_{ij} = \exp \left(-\frac{\|x_i - x_j\|^2}{4\epsilon} \right)$$

$$q_i = \sum_j W_{ij} \quad \Longrightarrow \quad \tilde{W}_{ij} = \frac{W_{ij}}{q_i q_j}$$

$$d_i = \sum_j \tilde{W}_{ij} \quad \Longrightarrow \quad P_{ij} = \frac{\tilde{W}_{ij}}{d_i}$$

$$L = \frac{P - I}{\epsilon}$$

Ambient Coordinates and the Metric Tensor

$$x = (c_1, c_2, \dots, c_D) \quad \text{where} \quad c : \mathbb{R}^D \rightarrow \mathbb{R}$$

$$\Gamma(c_a, c_b)(x) = \frac{1}{2} \sum_{m=1}^K P_{x \rightarrow m} (c_a(X_m) - c_a(x))(c_b(X_m) - c_b(x))$$

$$G_{ab} = \frac{1}{2} \sum_{m=1}^K P_{x \rightarrow m} (\Delta_m)_a (\Delta_m)_b$$

$$G = \frac{1}{2} \sum_{m=1}^K P_{x \rightarrow m} (\Delta_m \Delta_m^T)$$

Calculating the Tangent Space (Bypassing G)

$$\Delta_m = X_m - x$$

$$V_{\text{tan_raw}} = \sum_{m=1}^K P_{x \rightarrow m} (V \cdot \Delta_m) \Delta_m$$

$$V_{\text{tan}} = \hat{V}_{\text{tan_dir}} \left(V \cdot \hat{V}_{\text{tan_dir}} \right)$$

$$V_{\text{ortho}} = V - V_{\text{tan}}$$